

Quora Challenge NLP Deliverable

Team Members:

The work was distributed among the team members as follows:

- Miki: Focused on data cleaning and the basic simple solution/features.
- Jose: Investigated and coded string distance features (Jaccard Similarity and Length Differences).
- Lluc: Extracted advanced features, implemented the model training and evaluation notebooks.

Simple Solution:

Our simple solution relies on a basic logistic regression model trained on a single feature: Word Overlap. Limitations include that it ignores the order of words, synonyms, and context, often failing on questions that have the exact same words but completely different meanings (e.g., 'How do I read a book?' vs 'I read a book, how?'). This naive solution provides a low but useful baseline.

Improved Solution:

To improve upon the simple solution, we incorporated string distance metrics. Specifically, we implemented Jaccard Similarity and Length Difference between the strings. We extracted these features for both questions and concatenated them alongside the Word Overlap feature. This captures similarity more robustly, helping the model detect slight variations or significantly mismatched lengths that often correlate with non-duplicates.